

# Python Coding Best Practices

## Introduction

Python is valued for its readability and productivity. Well-structured Python code is easier to maintain, review, test, and extend. Coding standards help teams collaborate effectively and reduce defects over the lifetime of a project. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## Naming Conventions

Use descriptive names for variables, functions, classes, and modules. Follow PEP 8 naming conventions, such as snake\_case for functions and variables and PascalCase for classes. Avoid abbreviations that obscure intent. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## Functions

Functions should have a single responsibility and remain reasonably short. Minimize side effects, prefer returning values over modifying global state, and break complex logic into helper functions. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## Type Hints

Use type annotations to improve readability and enable static analysis tools such as mypy. Type hints help IDEs provide better autocompletion and make APIs easier to understand. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## Error Handling

Catch only expected exceptions, provide useful error messages, and avoid bare except blocks. Raise meaningful custom exceptions where appropriate and log failures with sufficient context. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## Testing

Develop automated unit and integration tests using pytest. Cover normal workflows, edge cases, invalid inputs, and regression scenarios. Continuous integration ensures tests run automatically. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## **Project Structure**

Organize projects into packages, separate business logic from infrastructure, centralize configuration, and isolate dependencies using virtual environments and pyproject.toml. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## **Performance**

Profile before optimizing. Prefer built-in data structures and algorithms. Use generators for large datasets and leverage concurrency only when appropriate. Practical examples, design

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## Documentation

Maintain module, class, and function docstrings. Keep README files updated and document setup instructions, architecture decisions, and public interfaces. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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## Conclusion

Following these practices leads to codebases that are easier to maintain, test, scale, and understand while reducing long-term technical debt. Practical examples, design considerations, common mistakes, and recommendations should be evaluated during development and code reviews. Teams benefit from documenting standards, mentoring new developers, and revisiting architectural decisions as systems evolve.

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